Sensor faults and actuator commands in a low-cost closed hydroponic system: a retrospective audit

MD. Sameer Iqbal Chowdhury[a], Mikdam-Al-Maad Ronoue[a], Md Asaduzzaman[b], Lafifa Jamal[a,*]

[a]*Department of Robotics and Mechatronics Engineering, University of Dhaka, Dhaka-1000, Bangladesh*
[b]*Department of Agriculture and Food Technology, Kyoto University of Advanced Sciences, Japan*

Abstract

Low-cost Internet of Things (IoT) controllers now regulate the nutrient solution and climate of closed hydroponic systems. In such controllers, a faulty sensor reading can directly trigger a dosing or injection command. As such, the effect of sensor faults on actuator commands needs to be measured, not only the faults themselves. We audited a 30-day log of 121,244 samples from our closed nutrient film technique (NFT) lettuce system. The log contained impossible pH values, prolonged full-scale readings and gaps covering 23% of the period. We replayed the reported setpoints and the executable actuator directions with a common 10-min cooldown. Under naive control, flagged readings supplied 19.1% of pH dosing commands and 10.2% of CO_2 injection commands. A physical-range check with 32 s persistence reduced these fractions to zero and 2.0%. The flags separate out-of-range values, bound readings and heuristic spikes, and are not independent fault diagnoses. Only 3% of valid air-temperature readings and 10% of valid water-temperature readings were in band, which indicates poor regulation with an unresolved cause. We also developed LiveGate, a forecast-consistency and sensor-liveness gate. Under a frozen retrospective protocol, it did not outperform simple baselines overall. A post-hoc comparison with run-length and bound-rejection gates exposed substantial availability costs. Learned liveness reduced commands from injected stuck readings by 79%, but useful-command coverage in those trials fell from 98% to 27%. The commands were reconstructed rather than

*Corresponding author.

observed, so actual dosing, prevented damage and crop benefits remain unmeasured.

Keywords: hydroponics, sensor faults, actuation, retrospective audit

1. Introduction

Closed hydroponic systems recirculate a nutrient solution whose pH, electrical conductivity (EC) and temperature must be held within narrow ranges [1, 2]. The surrounding air temperature, humidity and carbon dioxide (CO_2) concentration also shape growth in enclosed production [1, 2]. Automated controllers built from low-cost sensors, microcontrollers and single-board computers now regulate these quantities using dosing pumps, valves, chillers and coolers [3, 4].

In such controllers, a sensor reading is not only a data point, since it can trigger a physical action. A single spurious high pH reading can authorize acid dosing into the whole recirculating volume. The correction may then require dilution, counter-dosing and another mixing interval. Therefore, knowing how often a controller acts on flagged readings is as important as knowing how often the readings are flagged.

Studies of IoT hydroponic systems describe architectures, dashboards and crop outcomes, while sensor-fault studies characterize faults and their detection [5, 6, 7, 8]. Fault detection in hydroponics also predates the current IoT architectures [9], and SensorTalk connects agricultural fault diagnosis to the relationships between sensors and actuators [10]. Our question is narrower. We ask how flagged readings change the number of executable commands under a specified controller, and how much useful authorization is lost when readings are rejected. Answering it requires accounting for actuator direction, persistence, cooldown and the age of held readings together.

In this paper, we examine this question using data from our deployed closed NFT lettuce system. The contributions of this paper are as follows:

1. We provide a descriptive inventory of invalid, spiking, repeated and missing readings in a 30-day hydroponic deployment.
2. We developed an executable-command replay that attributes each authorization to its source reading, and used it to compare 6 gating rules.
3. We report a frozen retrospective comparison of LiveGate with 14 alternatives, including its negative overall result. This is followed by a separate comparison with simple run-length and physical-bound gates.

4. We derive engineering implications for gating, instrumentation and temperature regulation, with explicit limits on physical and agronomic inference.

The remainder of this paper is organised as follows. Section 2 reviews related work, and Section 3 describes our system and its deployment. Section 4 presents the audit, the controlled benchmark and LiveGate. Section 5 reports the results, Section 6 discusses them, and Section 7 concludes the paper.

2. Related Work

2.1. IoT Control of Hydroponic Systems

Automated hydroponic control typically measures pH and EC in the nutrient tank and corrects them with acid, base, concentrate or water dosing, often with a mixing delay between doses [3]. IoT variants add networked logging and remote monitoring [4]. In these studies, reliability is usually discussed in terms of communication and uptime. Evaluation focuses on whether the controlled variables stay near their setpoints, and on crop performance. However, how often the controller acts on readings that were wrong is rarely reported.

2.2. Sensor Faults and Their Detection

Field sensor deployments exhibit outliers, spikes, stuck-at values, offsets, drift and missing data [5, 6]. Data quality in IoT pipelines has also been reviewed systematically [7]. Reviews of agricultural IoT report similar fault classes, along with a growing set of model-based and learning-based detectors [8]. Classical detectors include:

1. Robust outlier filters [11]
2. Sequential change detectors such as the Cumulative Sum (CUSUM) [12, 13]
3. Exponentially Weighted Moving Averages (EWMA) [14]
4. State estimation with the Kalman filter [15]
5. Density-based novelty detection such as the Local Outlier Factor (LOF) [16]

A survey of model-based fault diagnosis is given in [17]. However, range and local-change tests can accept a plausible constant trace without independent evidence of a change in the process. Persistence-based confirmation can then reinforce an erroneous authorization.

2.3. Conformal Calibration

Split conformal prediction converts a predictor into a set or threshold with a finite-sample coverage guarantee under exchangeability [18, 19]. In our gate, its quantile rule provides a nominal calibration parameter. We make no formal coverage guarantee for these dependent sensor time series. As such, the nominal parameter is not a guaranteed rejection rate for clean readings, and performance is assessed empirically.

2.4. Diagnosis and Consequences in Controlled Environments

Ferentinos et al. [9] designed and developed neural-network detectors for mechanical, sensor and biological faults in deep-trough hydroponics. Their networks already use control information, since actuator control signals are among the inputs. Ferentinos and Albright [20] also proposed a real-time supervisor for the same setting. Lin et al. [10] designed SensorTalk, which temporarily feeds the readings of a standard sensor to the actuators during calibration. More recently, Karimzadeh et al. [21] addressed EC and pH fault diagnosis in hydroponic systems.

In controlled-environment agriculture, Munser et al. [22] used parity residuals and observers built on identified process relationships to isolate faults, subject to model and fault assumptions. The consequences of sensor errors also have a precedent. Bontsema et al. [23] used reference-measured greenhouse sensor errors to simulate their effects on production and energy use.

These diagnosis studies require different information and endpoints from retrospective command attribution. Thus, classification accuracy cannot be ranked against command coverage across different datasets. Our contribution is a deployment-specific authorization audit with explicit source provenance and executable directions. It is not a new general principle of agricultural fault diagnosis, nor evidence of measured physical consequences.

3. System and Deployment

In this section, we describe our hydroponic system, the controller logic replayed by the audit and the deployment that produced the log.

3.1. Platform

Our system is a closed NFT system for leaf lettuce, which consists of a Growth Module (GM) and a Nutrient Solution Container Module (NSCM). Figure 1 shows the sensing-to-actuation path of the system, along with the

deployed system itself. Table 1 lists the controlled quantities, their setpoint bands and the directions that the installed actuators can execute.

The GM measures air temperature and humidity with a DHT22 sensor, and CO_2 with an MH-Z19C non-dispersive infrared (NDIR) sensor whose range ends at 5000 ppm. CO_2 is raised by a normally closed solenoid valve on a CO_2 cylinder. Air temperature is lowered by a thermoelectric cooler, and LED grow lights provide the photoperiod.

The NSCM measures pH with an Atlas Scientific analog pH kit and EC with an Atlas Scientific Mini Conductivity K 1.0 kit. It also measures water temperature, and water level with a JSN-SR04T ultrasonic sensor. pH is corrected by acid or base dosing, EC by concentrate or distilled-water dosing, and water temperature by a refrigerative chiller. A pump circulates the solution continuously, and an aerator runs throughout.

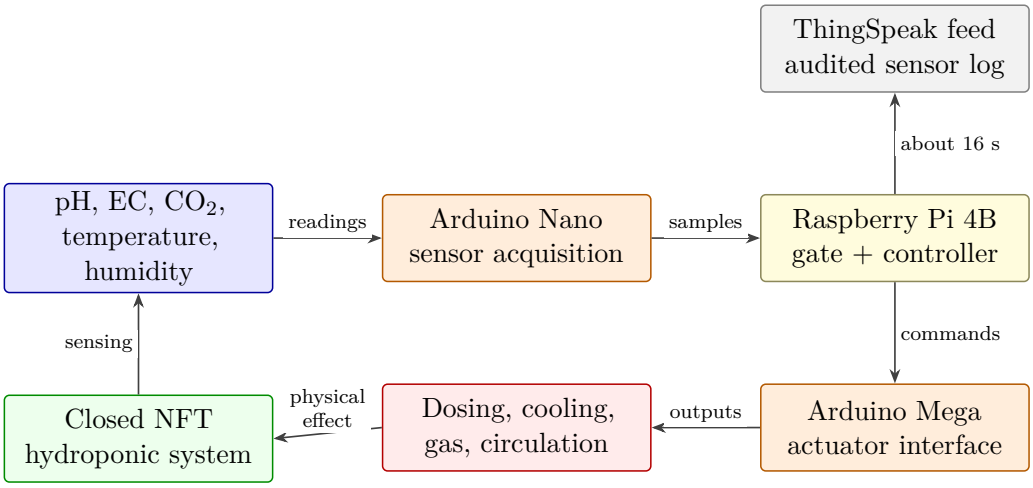
The sensors are read by an Arduino Nano, which applies median filtering, and the actuators are driven by an Arduino Mega. A Raspberry Pi 4 Model B runs the controller and uploads the readings to a ThingSpeak channel at intervals of approximately 16 s. However, actuator commands and relay states were not logged. As such, every command in this paper is reconstructed from the sensor log.

Table 1: Controlled quantities, deployed setpoint bands and executable actuator directions. Humidity and water level were monitored but not actuated.

Quantity	Setpoint band	Actuator	Executable direction
pH	5.8–6.5	Acid / base dosing	Both
EC	1000–1500 $\mu S\,cm^{-1}$	Water / concentrate dosing	Both
CO_2	1900–2100 ppm	Solenoid valve on cylinder	Raise only
Water temperature	18–22 °C	Refrigerative chiller	Lower only
Air temperature	20–26 °C	Thermoelectric cooler	Lower only

3.2. Replay Control and the Legacy-Gate Proxy

The description of our system specifies a correction whenever a reading leaves its setpoint band, followed by a 10-min mixing lockout after pH or EC



Command and relay states were not logged.



Figure 1: Sensing-to-actuation path (top) and the deployed NFT system (bottom). Sensor readings were logged, while command and relay states were not recorded.

dosing. The replay uses this interval as a common cooldown for all 5 channels. Its extension to CO_2 injection and cooling is a comparison assumption, not verified deployment timing. In the first growing season, the controller acted on logged readings without any further gating.

For the second season, we added a sensing rule that accepts a reading when it differs from the last trusted value by at most $(1+\beta)\xi$. The rule resets after 50 consecutive rejections. The description also specifies $\alpha\xi$ actuation bands around the setpoint limits and a repeatability count ψ. However, the band notation does not uniquely specify the executable firmware behavior. This led us to build a legacy-gate proxy, which works as follows:

1. It retains the sensing rule and its reset after 50 consecutive rejections.
2. It omits the α bands.
3. It requires 16ψ seconds of out-of-band persistence of an age-eligible held estimate in the common controller.
4. It re-estimates ξ and ψ from the development data, since their original first-season inputs are unavailable.

The published multipliers β were 0.0002 (EC), 0.0005 (pH), 0.0022 (air temperature), 0.0009 (water temperature) and 0.0010 (CO_2). Thus, this arm is an approximate comparison and not a demonstrated reconstruction of the firmware.

3.3. Deployment

The analysed deployment ran from 27 February to 28 March 2024 in Dhaka, Bangladesh. The ambient temperature was approximately 25–36 °C, which is off-season for lettuce. We grew 7 seedlings per treatment in 3 setups:

1. Our automated NFT system
2. Pots with soil and compost
3. A manually managed Deep Water Culture (DWC) system

4. Methods

In this section, we describe the data, the fault inventory, the replay audit and its sensitivity grid, the controlled benchmark and LiveGate.

4.1. Data

The log contains 121,244 rows with a median interval of 16 s. The analysis uses the 5 actuated quantities, along with humidity for the fault inventory. Water level is excluded, since its raw values were never calibrated to a physical level. The timestamps are split chronologically into 3 blocks: development (first 40%), validation (next 30%) and evaluation (last 30%).

4.2. Fault Inventory

For each channel, we counted 5 types of descriptive flags:

1. **Out-of-range readings:** Readings outside the physical range of the channel. The ranges are 0–14 for pH, 0–5000 $\mu S\,cm^{-1}$ for EC, 250–5000 ppm for CO_2, −10–60 °C for air temperature, 0–50 °C for water temperature and 0–100% for humidity.
2. **Isolated spikes:** In-range readings whose deviation from a centred 9-sample median exceeds 8 robust standard deviations.
3. **Flat-lines:** Runs of identical readings lasting at least 1 h without a logging gap.
4. **Full-scale readings:** The fraction of readings at the in-range maximum of the channel.
5. **Logging gaps:** Intervals between readings longer than 120 s.

These are descriptive definitions and not ground-truth labels.

4.3. Fault-to-Actuation Audit

The audit replays the reported setpoint bands and the common 10-min cooldown on the unmodified log. We compared 6 gating arms:

1. A naive threshold on every logged reading, as in the first season.
2. A physical-range check.
3. A range check with 32 s of persistence before acting.
4. The approximate legacy-gate proxy described in Section 3.2.
5. Arm 3 with the LiveGate liveness test added (Section 4.6).
6. Arm 3 with rejection of readings exactly at a declared physical bound.

In every arm, the last accepted value may be held for at most 64 s, and a logging gap longer than 120 s invalidates it. Directions unavailable to the installed actuator are suppressed before persistence and cooldown are

evaluated. Thus, an unavailable direction cannot delay a later executable command.

Each command is attributed to the reading that supplied its held estimate. We report 3 disjoint flag categories: out-of-range values, readings exactly at a declared bound, and in-range isolated spikes that are not already at a bound. A bound reading may represent a true excursion beyond the range of the sensor, and a spike may be a legitimate transient. As such, neither is an independently confirmed sensor fault.

The replay uses a fixed historical background, so its commands do not change subsequent readings. It therefore measures differences in authorization, not actual dosing, prevented damage or crop effects. Band occupancy is the fraction of logged readings, within range and away from declared bounds, that fall inside the setpoint band. It is not a duration-weighted estimate for the missing periods.

4.4. Post-hoc Timing and Flag Sensitivity

The original audit reported a single timing setting. After reproducing it, we declared a separate sensitivity grid before executing it. The grid varied the following settings:

1. Persistence of 16, 32, 64 or 96 s, with the naive and range-only controls kept at zero persistence.
2. Maximum held age of 32, 64 or 128 s.
3. Cooldown of 300, 600 or 1200 s.
4. Spike-attribution cutoffs of 6, 8 or 10 robust scales. These descriptive cutoffs did not alter the online gate.

The 3 arms, 5 channels and 3 spike definitions produced 1620 reported cells, including repeated control cells, and every cell was retained. The original setting had to reproduce before the full grid was run. These deterministic settings are sensitivity cases from one deployment and not independent replicates. Authorization retention is the command count relative to naive replay at the same age and cooldown. It is not the coverage of independently verified necessary actions.

4.5. Controlled Benchmark

Since the log has no actuator records or reference instruments, faults that cannot be identified from the log alone were studied by controlled injection

into real backgrounds. We drew 8 non-overlapping, gap-free 240-sample windows from each of the validation and evaluation blocks. In each window, 1 of 12 scenarios is applied to 1 channel after a 60-sample warm-up. The scenarios are as follows:

1. A clean window with no injection.
2. Positive or negative spikes.
3. Positive or negative biases lasting 30 samples.
4. Positive or negative drifts lasting 30 samples.
5. A stuck value lasting 30 samples.
6. Saturation, with 30 samples pinned at the declared physical maximum.
7. A dropout lasting 30 samples.
8. Positive or negative legitimate level shifts, which also change the reference background.

The magnitudes are 0.5 for pH, 100 $\mu S\,cm^{-1}$ for EC, 2 °C for air temperature, 1 °C for water temperature and 500 ppm for CO_2.

Every method produces an accepted or rejected decision for each reading. A common downstream controller holds the last accepted value for up to a maximum age (64 or 320 s), requires out-of-band persistence (0, 32 or 96 s) and applies the 10-min cooldown. Reference opportunities are defined from the uncorrupted background, with a fixed 32 s persistence and the same cooldown. An authorization matches an opportunity if it occurs in the same out-of-band episode and direction within 180 s. The 2 main metrics are defined as follows:

1. **Coverage:** The fraction of reference opportunities that are matched.
2. **Undesirable authorizations:** Commands issued when the background is in band, or in the opposite direction.

The benchmark retains the idealized two-direction controller of the frozen protocol for every channel. Thus, its counts are distinct from the executable-command audit.

Two secondary metrics separate authorizations that occur during injected-fault samples from authorizations whose last accepted source reading was fault-labelled. The former is the temporal-coincidence metric of the original protocol, which does not establish that a corrupted reading was used. The latter follows the provenance of the held estimate, including commands issued after a fault ends. However, neither metric alone establishes an undesirable direction relative to the background.

The detector setting, persistence and maximum age of each method are selected on the validation windows at 3 coverage targets (0.5, 0.8 and 0.95). Among the settings meeting a target, the one with the fewest undesirable authorizations per trial is locked, and the evaluation windows are then scored once. The 14 alternatives to full LiveGate are:

1. Persistence only
2. Hampel filter
3. CUSUM
4. LOF
5. Kalman filter, with noise scaled per channel from the development data
6. EWMA
7. Moving median
8. Generalized Likelihood Ratio (GLR)

9–11. 3 AASVR-R supervisor variants (measurement supervisor, hold-only and without CUSUM)

12. Always-deny

13–14. 2 LiveGate ablations

Each split contains 480 channel–scenario trials derived from 8 source windows. The evaluation windows span 7 source days, not 480 independent deployments. Paired 95% percentile intervals resample those days jointly across sensors and scenarios with 1000 bootstrap replicates [24]. However, 7 clusters provide limited information about between-day variability. The component intervals are descriptive and have no multiplicity adjustment, and the settings remain fixed during resampling.

4.6. LiveGate

We developed LiveGate to accept a reading y_t for actuation only if it is consistent with a learned forecast and the sensor appears alive. It comprises a consistency test, a liveness test and a decision rule, which are described below.

Consistency. Let ℓ_t be the mean of the last $K = 5$ accepted readings, and τ_t the difference between the newest and oldest of them. The forecast is given by Equation 1:

$$\hat{y}_t = \ell_t + \mathbf{w}^\top \boldsymbol{\phi}_t + b, \qquad \boldsymbol{\phi}_t = \Big[\tau_t,\ \sin(2\pi h_t/24),\ \cos(2\pi h_t/24),\ \mathbf{z}_t\Big] \quad (1)$$

h_t is the hour of day, and $\mathbf{z}_t$ contains the other 4 actuated channels, winsorized at their development 1st and 99th percentiles and centred at their medians. The weights are fitted by Huber regression [25] of $y_t - \ell_t$ on the first 75% of the development block, using only in-range, non-spike readings. The score is $s_t = |y_t - \hat{y}_t|/\sigma$, where σ is 1.4826 times the median absolute training residual. The threshold is the split-conformal quantile given by Equation 2:

$$q_\varepsilon = s_{(\lceil (n+1)(1-\varepsilon) \rceil)} \tag{2}$$

$s_{(k)}$ is the k-th smallest of the n clean scores in the last 25% of the development block.

Liveness. Let r_t be the length of the current run of identical readings. The survival function is estimated from the N runs of filtered readings in the first 75% of development, excluding readings at a declared physical bound. The smoothed survival function is given by Equation 3:

$$S(r) = \frac{|\{\text{runs} \geq r\}| + 1}{N + 1} \tag{3}$$

$S(r)$ is the smoothed fraction of training runs lasting at least r readings. The sensor is considered alive while $S(r_t) \geq \varepsilon$. The frozen implementation counts training runs after filtering and does not reset its liveness counter at logging gaps. The controlled comparison uses gap-free windows, but the whole-log liveness results remain subject to this gap-handling limitation.

Decision. A reading is accepted if it is within the physical range, the sensor is alive and $s_t \leq q_\varepsilon$. A live but surprising reading starts a candidate level. The trusted state then re-anchors after the number of mutually consistent readings given by Equation 4:

$$m(s_t) = \text{clip}\Big(1 + \lceil \log_2(s_t/q_\varepsilon) \rceil,\ 2,\ 8\Big) \tag{4}$$

$m(s_t)$ is the number of confirming readings required, so larger surprises require more confirmation. The single parameter $\varepsilon \in \{0.002, 0.01, 0.05\}$ sets both the consistency and liveness thresholds. Thus, it takes the place of the threshold multipliers of the baselines.

Protocol. Before running LiveGate on the measured log, we committed the method, the benchmark and the decision rules to the project repository. LiveGate would be called best at a coverage target only if it had the lowest evaluation undesirable rate among the methods feasible at that target, with coverage within 0.02 of the best feasible method. It would be called best overall only if this held at every target where it was feasible. Our initial forecast was anchored on the last accepted value. As we ran the synthetic unit tests, we noticed cascading rejections, which led us to anchor the forecast on a 5-sample level instead. This was the only design change, and it was made before the measured log was used. However, the data had been analysed in earlier studies, so the evaluation is retrospective rather than on untouched data.

5. Results

In this section, we present the faults found in the log, the observed regulation, the audit and its sensitivity, and the controlled benchmark.

5.1. Faults in the Deployment Log

Table 2 lists the fault inventory of the 30-day log, and Figure 2 shows the daily flags. Flagged readings were frequent and channel-specific. Our observations are as follows:

1. pH produced 464 physically impossible readings, with values up to 67,101. This is consistent with corrupted analog or serial transfers, although the location of the fault cannot be diagnosed from the log.
2. CO_2 produced 819 out-of-range readings, with values up to 200,850 ppm, and 961 isolated spike flags. Furthermore, 2.3% of its samples were exactly at the 5000 ppm full scale, including one episode lasting 8.5 h.
3. Humidity read 99.9% for 99.97% of samples. Saturation, sustained near-condensation conditions and a stuck or corrupted channel cannot be distinguished without independent measurements.
4. EC and both temperature channels had fewer flags. However, this does not establish their calibration accuracy.
5. All channels shared 194 logging gaps longer than 120 s. The longest gap lasted 9.0 h, and together the gaps covered 23% of the deployment.

Table 2: Fault inventory of the 30-day deployment log, with 121,244 readings per channel. Spikes are in-range readings more than 8 robust standard deviations from a centred 9-sample median.

Channel	Out of range	Maximum out-of-range	Spikes	Flat-lines ≥1 h	Longest (h)
pH	464	67,101	76	0	–
EC	0	–	14	0	–
CO_2	819	200,850	961	3	8.5
Air temperature	1	99.9	17	1	1.4
Water temperature	0	–	39	7	1.5
Humidity	0	–	10	79	35.2

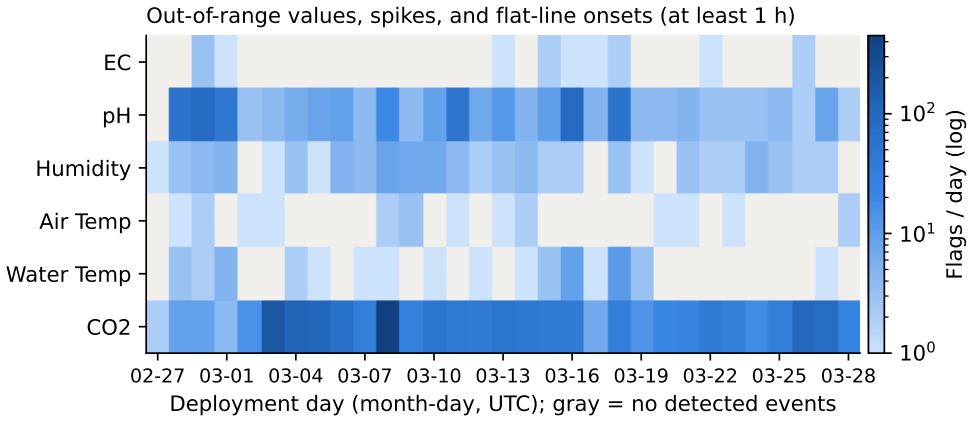


Figure 2: Daily descriptive flags: out-of-range readings, isolated spikes and onsets of flat-lines lasting at least 1 h. These flags are not independent diagnoses of sensor faults.

Figure 3 shows the longest CO_2 full-scale episode. It was preceded by a steady rise from approximately 2000 to 5000 ppm over 2 h. The true concentration may have exceeded the sensor range, such as through a valve that failed open. However, reference and valve-state measurements are absent. If the sensor was functioning and clipped at full scale, the reading gives a lower bound. If it was stuck, even that interpretation is unavailable. The replayed liveness test rejected the pinned readings within 4 samples.

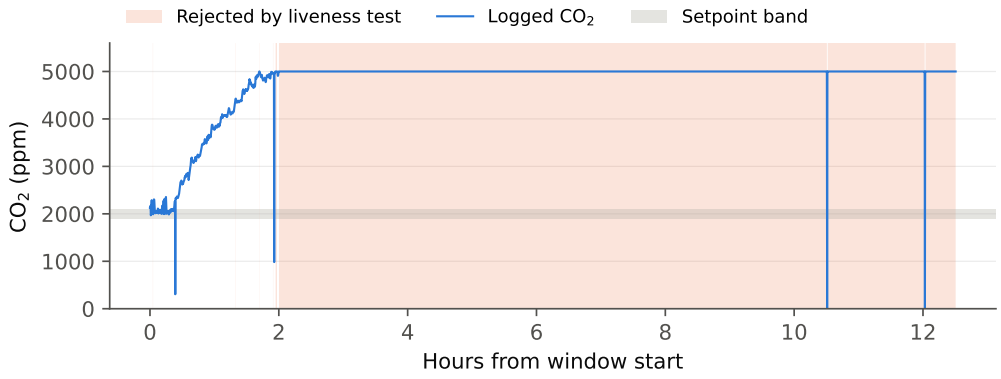


Figure 3: Longest CO_2 full-scale episode, with readings rejected by the replayed liveness test ($\varepsilon = 0.05$) shaded. Downward excursions are spike candidates, and their physical cause is unverified.

5.2. Observed Regulation

Figure 4 shows the fraction of valid readings below, within and above each setpoint band. Among valid logged readings, 88% of pH, 84% of EC and 47% of CO_2 readings were in band. Only 3% of valid air-temperature readings were within 20–26 °C, with a median of 28.6 °C. Similarly, 10% of valid water-temperature readings were within 18–22 °C, with a median of 24.8 °C. These sample proportions do not describe the unobserved 23% of the deployment.

The traces indicate poor temperature regulation, which is consistent with insufficient cooling capacity under hot-season conditions. However, actuator failure, control problems and calibration error remain possible. Without command, relay, reference and capacity measurements, the log cannot identify the cause or the actual cooling duty.

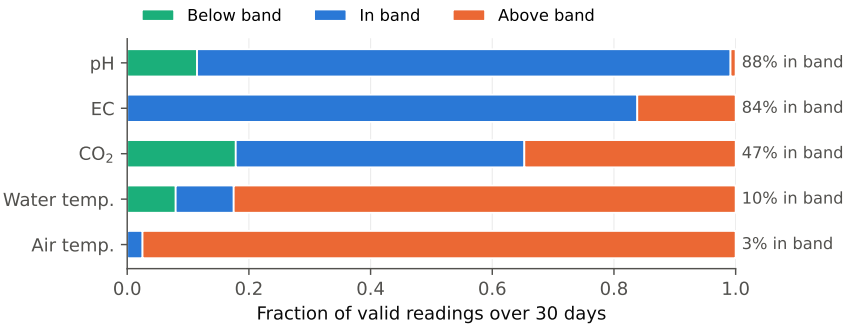


Figure 4: Fraction of valid readings below, within and above each setpoint band over the deployment

5.3. Fault-to-Actuation Audit

Table 3 lists the executable replay commands under each gating arm, and Figure 5 shows the share of them that used flagged readings. Our observations are as follows:

1. A naive controller replay issued 655 pH dosing commands, of which 125 (19.1%) used flagged readings. These comprise 89 out-of-range values, 25 bound readings and 11 heuristic spikes.
2. Naive replay also issued 2149 CO_2 injection commands, of which 219 (10.2%) used flagged readings. These comprise 74 out-of-range values and 145 spikes.
3. A range check left 40 flagged pH commands, made up of 28 bound readings and 12 spikes.
4. Adding 32 s of persistence removed all flagged pH and EC commands. It also reduced flagged CO_2 injections to 9 of 456 (2.0%), all 9 attributed to heuristic spikes. This shows that short confirmation removed most flagged-source commands at the original timing.
5. Liveness and bound rejection produced the same executable CO_2 counts as range plus persistence. This is because high full-scale readings do not request injection in a raise-only loop.

Our initial audit discarded unavailable directions only after they had consumed cooldown. Upon investigation, we learned that this implementation error allowed an unavailable direction to delay later executable commands. This led us to enforce the directions before authorization. The correction

changed the naive CO_2 counts from 1010/68 total/flagged to 2149/219, and the range-plus-persistence counts from 441/7 to 456/9. The two-sided pH and EC results were unchanged. The separate frozen controlled benchmark deliberately remains bidirectional.

The legacy-gate proxy issued 43–72% as many commands as the range-and-persistence arm, depending on the channel. This was because its repeatability requirement corresponded to approximately 2–7 min of confirmation. Nevertheless, it authorized 13 CO_2 injections using out-of-range readings, since its reset accepts the current sensor value after 50 consecutive rejections. These are proxy outputs and not evidence that the deployed firmware issued those commands.

Table 3: Executable replay commands and commands using flagged source readings. Flags are out-of-range, declared-bound or heuristic-spike readings. Tot.: total commands. Flag.: commands using flagged readings. No actual command execution was logged.

	pH dosing		EC dosing		CO_2 injection	
Gating arm	Tot.	Flag.	Tot.	Flag.	Tot.	Flag.
Naive threshold	655	125	535	6	2149	219
Range check	593	40	535	6	2105	148
Range + persistence	417	**0**	526	**0**	456	9
Legacy-gate proxy	216	**0**	227	**0**	330	13
Range + persistence + liveness	417	**0**	526	**0**	456	9
Range + persistence + bound	417	**0**	526	**0**	456	9

The liveness test reduced the replayed air-temperature cooling commands by 10% (3215 to 2891) and the water-temperature commands by 9% (2698 to 2446). Coarsely quantized channels can repeat values during legitimate operation. Thus, these reductions are a cost in authorization, not observed reductions in cooling or evidence of physical benefit.

5.4. Sensitivity of the Audit Finding

Table 4 lists the ranges across the complete post-hoc grid for range plus persistence. The grid falsified our stronger prediction that this arm would remove every flagged pH authorization across timings. There were 0 to 3 flagged pH commands per setting, with nonzero counts in 54 of 108 pH attribution cells. With 32 s persistence and 64 s maximum age, shortening the cooldown from 600 to 300 s admitted 1 bound-source command at the original 8-scale spike cutoff. In contrast, EC remained at zero throughout.

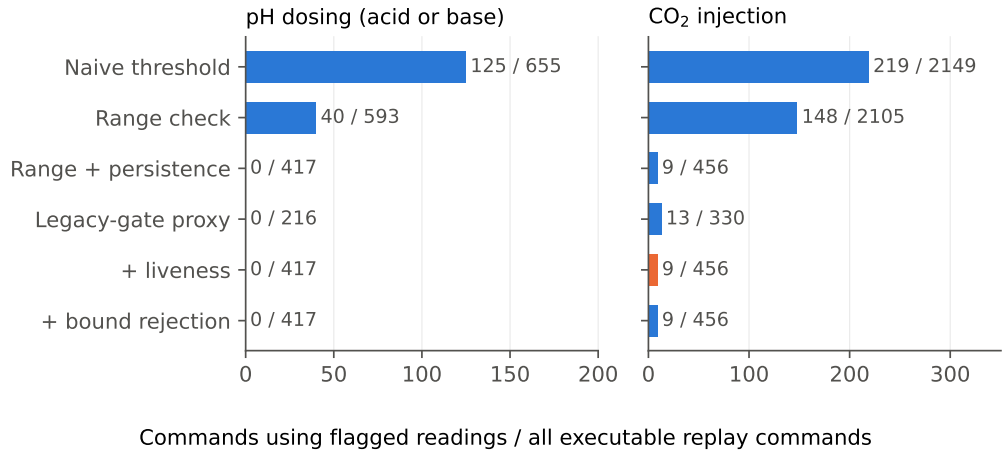


Figure 5: Executable replay commands using flagged readings, relative to all executable commands under each arm. The flags do not establish physical harm.

Relative to naive replay at the same timing and spike definition, flagged counts fell by 98.1–100% for pH and by 87.1–99.7% for CO_2. These are whole-log descriptive reductions and not estimates of prevented harm. Rare air-temperature flags did not decrease uniformly, and in one comparison a count rose from 1 to 2 as authorizations were rescheduled. As such, the useful scope of the finding is a measured-log authorization tradeoff with explicit timing boundaries. It is not a guarantee that short persistence eliminates faulty decisions.

Table 4: Ranges across the complete post-hoc timing and spike-attribution grid for range plus persistence. Flagged counts use disjoint out-of-range, bound and spike categories. Each row summarizes 108 attribution cells from one deployment. Retention is relative to naive command counts at the same timing, and does not establish useful-action coverage.

Channel	Total commands	Flagged commands	Retention (%)
pH	207–829	0–3	50.7–74.7
EC	267–1032	0	97.4–99.1
CO_2	121–1576	1–68	9.8–53.2
Air temperature	1642–6275	0–2	98.5–99.8
Water temperature	1368–5288	0	98.7–99.9

5.5. Controlled Benchmark and the Locked LiveGate Result

Table 5 lists the locked evaluation results, and Figure 6 shows all validation operating points. No method reached the 0.95 coverage target on validation. At the 0.5 and 0.8 targets, LiveGate did not meet the frozen criterion for the best method. Our observations are as follows:

1. At 0.5, LOF had the fewest undesirable authorizations, with 0.075 per trial against 0.135 for LiveGate (paired difference 95% interval −0.148 to −0.004). However, its coverage was much lower, at 0.357.
2. At 0.8, persistence only reached a higher coverage (0.886) with fewer undesirable authorizations (0.117) than LiveGate (0.790 and 0.135).
3. The liveness-only ablation had the lowest undesirable rate at 0.8 (0.104), but its evaluation coverage fell to 0.663.
4. The AASVR-R measurement supervisor did not reach the 0.5 target.

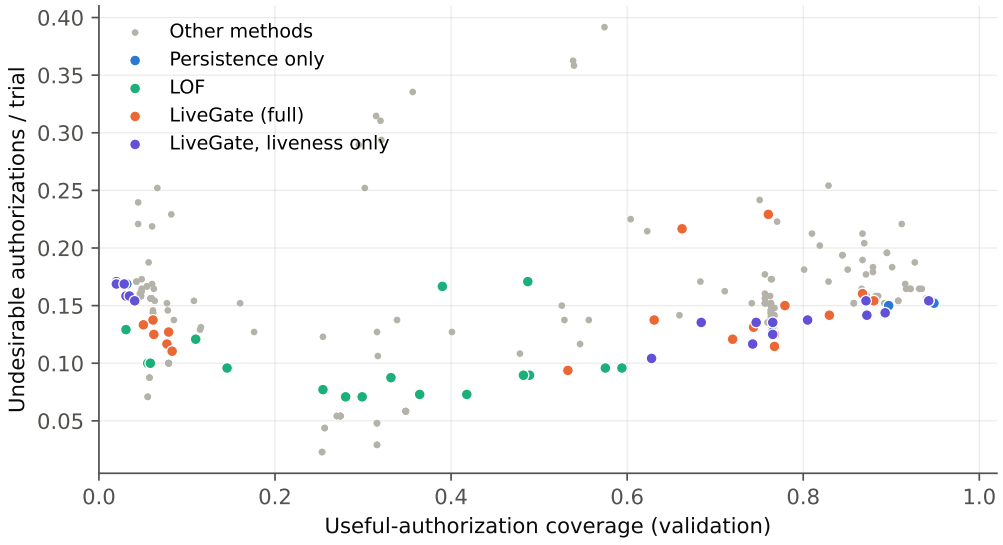


Figure 6: Validation operating points of all methods and settings. Points towards the lower right authorize more of the needed commands while issuing fewer unneeded ones.

The forecast-only ablation rejected 3.2% of uncorrupted readings at the 0.8 target, against none for persistence only. It also had more undesirable authorizations, with 0.150 against 0.117 per trial. Thus, its extra rejection did not improve the aggregate authorization tradeoff in this benchmark. Since a

Table 5: Locked evaluation at the 180 s deadline. Settings were selected on validation, and validation-infeasible methods are omitted. A dagger marks evaluation coverage below the corresponding target. Bold values are column extrema, not proof of an overall winner.

Target	Method	Coverage	Undesirable / trial
0.5	LOF	0.357†	**0.075**
	LiveGate, liveness only	0.560	0.081
	Kalman	0.594	0.098
	GLR	0.760	0.100
	Persistence only	0.795	0.110
	CUSUM	0.794	0.110
	Hampel / moving median	0.795	0.121
	EWMA	**0.801**	0.121
	LiveGate	0.505	0.135
	LiveGate, forecast only	0.704	0.163
	AASVR-R hold only	0.721	0.233
0.8	LiveGate, liveness only	0.663†	**0.104**
	Persistence only	0.886	0.117
	CUSUM	0.862	0.121
	LiveGate	0.790†	0.135
	Hampel / moving median	0.848	0.150
	EWMA	0.857	0.150
	LiveGate, forecast only	0.837	0.150
	Kalman	**0.910**	0.158
	GLR	0.854	0.165

fault and a legitimate change can produce the same local trace, any additional context would have to be informative to distinguish them.

5.6. Stuck and Saturated Sensors

Figure 7 shows the authorizations that occurred during injected-fault samples. On the original temporal-coincidence metric, enabling liveness reduced the authorizations during stuck and saturated intervals. Our observations at the 0.8 validation target are as follows:

1. During stuck samples, persistence only authorized 0.725 times per trial, compared with 0.225 for liveness only (69% fewer) and 0.500 for full LiveGate.
2. During saturation, the corresponding rates were 0.975, 0.675 (31% fewer) and 0.650. Kalman reached 0.850, and the other selected comparators matched persistence on this metric.

However, these reductions do not by themselves show that fewer commands used corrupted source readings. They also come with a reduced overall coverage of 0.663 for liveness only, against 0.886 for persistence. The following analysis adds paired intervals, held-source attribution and direct simple comparators.

5.7. Exploratory Post-hoc Variant

After the locked run, we noticed that the forecast scale had degenerated on the coarsely quantized temperature channels. This led us to test an exploratory variant that floors the forecast scale at the reading resolution of each sensor. All 66 prior evaluated rows reproduced exactly, and the selected settings and results of the variant matched locked LiveGate at both feasible targets. Upon investigation, we learned that the conformal threshold is calibrated on scores divided by the same scale, so rescaling moves the scores and the threshold together. As such, the variant did not change the authorization tradeoff of the forecast component, and we retain it as a disclosed null result.

5.8. Post-hoc Direct-Baseline Comparison

After the locked comparison, we recognised that the liveness and saturation results had not been compared against the simplest rules aimed at the same faults. This led us to rerun the original methods with 2 direct comparators added:

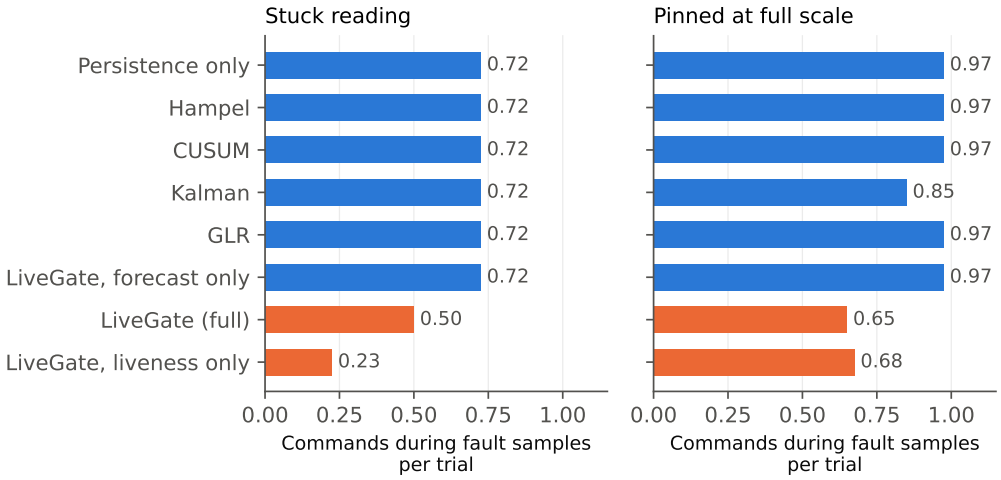


Figure 7: Authorizations occurring during injected-fault samples, per trial (locked evaluation, 0.8 validation target). This temporal metric does not identify the source of a held estimate. Liveness arms are highlighted.

1. **Fixed repeat cap:** The gate accepts the first R identical in-range readings and rejects later repeats. It resets after a change, an invalid reading or a logging gap. The cap $R \in \{2, 4, 8, 16, 32, 64\}$ is shared across channels, which gives it a larger grid than the 3 values of LiveGate.
2. **Physical-bound rejection:** The gate rejects any reading exactly at either physical bound of the sensor.

Both comparators use the original timing grid and are selected on the validation windows only. All original evaluation summary rows and selections reproduced, and the frozen artifacts remained unchanged. The follow-up is post-hoc, since the data and the original results had already been examined. Table 6 lists the selected operating points at the 0.8 target.

Physical-bound rejection issued 0.000 commands per saturation trial from an injected-fault source, while its useful-command coverage in those trials was 0.073. Thus, learned liveness was not the only rule that suppressed saturated source readings. At the 0.8 target, both the fixed-cap gate and the physical-bound gate improved overall coverage and the undesirable-command rate relative to full LiveGate. Physical-bound rejection also improved both relative to learned liveness only, although its evaluation coverage (0.798) fell slightly

Table 6: Post-hoc evaluation at the 0.8 validation coverage target. A dagger marks evaluation coverage below 0.8. Fault-source counts attribute each command to its last accepted reading, per trial. All counts are simulated authorizations.

Method	Coverage	Undesirable / trial	Fault-source commands Stuck	Saturation
Persistence only	0.886	0.117	0.725	0.975
LiveGate	0.790†	0.135	0.500	0.575
Learned liveness only	0.663†	0.104	0.150	0.675
Fixed repeat cap	0.849	0.117	0.675	0.975
Physical-bound rejection	0.798†	0.083	0.725	0.000

below 0.8. In comparison, persistence only reached a higher coverage than the fixed-cap gate at the same undesirable rate. These are point-estimate operating comparisons. They are not statistical superiority claims or evidence of field safety.

At the 0.8 target, validation selected a repeat cap of 64, 32 s persistence and 64 s maximum age for fixed repeat cap.

At the 0.8 target, validation selected 32 s persistence and 64 s maximum age for physical-bound rejection.

For stuck faults, learned liveness only reduced commands relative to persistence by 69.0 [54.3, 85.7]% during injected-fault samples, and by 79.3 [71.0, 88.9]% for commands using injected-fault source readings. The brackets give paired source-day 95% intervals.

However, useful-command coverage in the same stuck trials fell from 0.982 with persistence to 0.273 with learned liveness. As such, these reductions do not establish better fault handling at matched useful-command coverage.

For saturation faults, learned liveness only reduced commands relative to persistence by 30.8 [18.0, 46.2]% during injected-fault samples, and by 30.8 [18.0, 46.2]% for commands using injected-fault source readings. The brackets give paired source-day 95% intervals.

However, useful-command coverage in the same saturation trials fell from 0.918 with persistence to 0.409 with learned liveness. As such, these reductions do not establish better fault handling at matched useful-command coverage.

The 2 counters answer different questions, since a command during a fault interval can still use a held clean reading. Neither counter measures physical execution or damage. The estimates use 8 evaluation windows on 7 source-

day clusters. The settings remain fixed during resampling, and the secondary intervals have no multiplicity adjustment. Thus, the small number of days makes generalization uncertain. Table 7 lists the sensor-specific effects of learned liveness.

Table 7: Sensor-specific learned-liveness effects relative to persistence at target 0.8. Differences are method minus persistence. Fault-source differences are commands per trial; brackets give paired source-day 95% intervals. Coverage differences pool all scenarios.

Sensor	Coverage difference	Stuck difference [95% CI]	Saturation difference [95% CI]
EC	-0.167	-0.750 [-1.000, -0.400]	-0.750 [-1.000, -0.400]
pH	-0.231	-0.375 [-0.750, -0.091]	-0.375 [-0.750, -0.091]
Air temp.	-0.458	-0.750 [-1.000, -0.499]	0.000 [0.000, 0.000]
Water temp.	-0.067	-0.250 [-0.500, 0.000]	0.000 [0.000, 0.000]
CO_2	-0.071	-0.750 [-1.000, -0.429]	-0.375 [-0.714, 0.000]

At a fixed ε, the learned rule is itself a sensor-specific repeat threshold. Comparison with a global fixed cap therefore does not establish superiority over every per-sensor threshold. Any benefit must be weighed against the useful authorizations lost by rejecting legitimate repeated readings.

5.9. Crop Outcome in the Same Season

Table 8 lists the harvest of the analysed season. The NFT plants produced a mean aerial fresh mass of 264.8 g (SD 35.0, $n = 7$). In comparison, the soil and manual DWC plants produced 68.2 g (SD 19.5) and 109.5 g (SD 23.3). However, these treatments differ at once in growing medium, enclosure, CO_2 enrichment, cooling, nutrient management and automation. As such, the comparison describes the deployment context and does not isolate the effect of automation or of any gating rule.

Table 8: Harvest of the season analysed (mean ± SD, $n = 7$ plants per treatment). FMAP and DMAP: fresh and dry mass of the aerial part. TPL: total plant length. TNL: total number of leaves. NL10: leaves longer than 10 cm.

Trait	NFT (automated)	Soil	Manual DWC
FMAP (g)	264.81 ± 35.02	68.24 ± 19.46	109.49 ± 23.31
DMAP (g)	33.19 ± 8.24	8.63 ± 2.51	13.85 ± 2.70
TPL (cm)	72.52 ± 11.61	19.46 ± 3.54	20.12 ± 2.04
TNL	20.29 ± 2.87	9.86 ± 1.68	11.86 ± 1.95
NL10	13.00 ± 2.00	3.00 ± 1.15	7.57 ± 1.51

6. Discussion

6.1. Which Faults Matter for Actuation

The audit separates conspicuous flags from the flags that supply executable commands. Under naive replay, 1 in 5 pH authorizations used an out-of-range, bound or spike-flagged reading. A range check with 32 s persistence removed these flagged-source authorizations in this log, while reducing total pH commands from 655 to 417. However, without reference measurements, this does not establish that every suppressed command was unnecessary.

High CO_2 full-scale readings requested no injection, since that loop can only raise the concentration. Conversely, a pinned low reading, or saturation in a two-sided controller, can support repeated authorizations subject to cooldown. Thus, the same sensor signature has different command consequences depending on the actuator direction and the supervisor state.

6.2. Why Simple Rules Competed Well

In this benchmark, a bias fault and a legitimate level shift can initially produce identical single-channel observations. This construction makes confirmation time a useful discriminator, and helps explain why persistence performed well among the tested settings. However, it does not establish an optimal frontier for other deployments, faults or informative multichannel models.

Repeated readings provide another signature, but normal quantization can produce the same pattern. At a fixed ε, the learned survival rule is itself a sensor-specific run-length threshold. Therefore, its value must be judged

against direct run-length and bound-rejection rules at comparable coverage, and not only against detectors designed for changing signals.

6.3. Design Guidelines

Based on our results, we suggest the following design checks:

1. Physical-range checks and short persistence should be evaluated with the actual actuator directions and cooldown. In this log, they removed all flagged pH and EC source commands.
2. Resets that eventually accept any value after repeated rejections should be avoided. The legacy-gate proxy admitted out-of-range CO_2 readings through such a reset.
3. Run-length and bound rejection should be compared against persistence at matched coverage before liveness gating is enabled. Otherwise, coarse sensor resolution can suppress useful cooling authorizations.
4. Sensor ranges should be selected with headroom, and humidity sensing should be tested near condensation. A pinned channel cannot resolve further changes.
5. Cooling capacity, actuator duty and reference temperatures should be measured before poor regulation is diagnosed or equipment is resized.
6. Commands, relay states and independent references should be logged alongside the readings. These measurements are needed to connect differences in authorization to physical execution and outcomes.

6.4. Limitations

As with any retrospective study, our audit leaves room for future improvement. Its main limitations are as follows:

1. The evidence comes from 1 system and 1 season of 30 days.
2. The audit counts authorizations under a specified common controller and an approximate legacy-gate proxy. The original firmware and the actual execution records are unavailable.
3. Spike labels are heuristic, bound values are ambiguous, and the missing intervals prevent full-season occupancy estimates.
4. The synthetic benchmark uses fixed historical backgrounds, an idealized bidirectional controller and only 7 evaluation-day clusters. Its intervals describe the resampling variability for those backgrounds and fixed settings, not prospective deployment uncertainty.

5. The data and time blocks had already been analysed, and the follow-up baselines were added after the frozen comparison. These baselines are explicitly post-hoc.
6. The large CO_2 full-scale episodes lie in the development block. They were excluded from liveness fitting by the declared-bound rule, but they were not unseen in time.
7. The crop comparison is confounded by simultaneous differences between the treatments.

A prospective deployment with recorded commands, relay states, independent references and a frozen configuration is needed to establish physical execution, avoided damage or crop benefits.

7. Conclusions

In this paper, we audited how sensor faults propagate into actuator commands in our low-cost closed NFT hydroponic system. Our audit links descriptive sensor flags to executable replay commands while respecting actuator direction and cooldown. Under naive replay, flagged sources supplied 19.1% of pH commands and 10.2% of CO_2 commands. Range checks with 32 s persistence reduced these to zero and 2.0% at the original timing. However, the sensitivity grid admitted up to 3 flagged pH commands while retaining zero EC flags, so the zero-count pH result is specific to the setting. The temperature traces establish poor observed regulation, although its cause remains unresolved.

LiveGate did not outperform simple rules overall. The post-hoc comparison showed that bound rejection removed saturated-source authorizations, while learned liveness reduced stuck-source authorizations at a substantial cost in coverage. Neither result establishes a general advantage at matched availability. Thus, the engineering contribution of this work is an explicit account of how reading validity, held-value age, actuator direction and supervisor timing jointly affect authorization. Actual execution and agronomic benefits remain unmeasured.

Data and Code Availability

The code, configurations, frozen and follow-up evaluation protocols, measured 30-day sensor log, harvest data and supporting results are publicly

available at https://github.com/MSIChowdhury/Design-of-an-IoT-enabled-Scalable-Closed-Hydroponics-System-with-Intelligent-Parameter-Control-and-P. A labelled synthetic stand-in is also provided for code-level smoke tests, and it is not used for any reported finding.

Declaration of Competing Interest

The authors declare no competing interests.

Declaration of Generative AI and AI-assisted Technologies

AI-assisted tools were used to help write analysis code, draft text and check consistency. The authors reviewed the content and are responsible for it.

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